

KAMPHAENG PHET, Thailand

WMO: 483800

Lat: 16.4867N

Lon: 99.5269E

Elev: 82

StdP: 100.35

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	16.1	17.7	11.8	8.7	22.9	13.4	9.7	23.7	3.4	29.3	3.1	28.7	0.1	280	0.305

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.9	38.2	25.7	37.1	25.8	36.1	25.6	28.1	33.5	27.7	32.8	27.3	32.3	1.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.6	30.2	26.4	22.2	29.8	26.2	21.8	29.3	90.7	33.7	88.7	33.0	87.3	32.4	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
3.3	2.6	2.2	DB	13.7	39.9	1.8	1.5	12.4	40.9	11.4	41.8	10.4	42.6	9.1	43.6	
			WB	12.7	28.9	1.8	0.8	11.4	29.5	10.4	30.0	9.4	30.4	8.1	31.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	25.3	27.3	29.3	31.0	30.2	29.4	28.8	28.6	28.5	28.2	27.1	25.2
	DBStd	2.35	2.09	1.88	1.80	1.76	1.95	1.17	1.17	0.92	1.02	1.16	1.59	2.28
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6667	476	486	598	629	627	583	583	578	556	565	514	472
	CDD18.3	3625	217	252	339	379	369	333	324	319	306	307	264	214
	CDH23.3	40766	1987	2774	4249	5224	4621	3886	3611	3442	3265	3261	2578	1868
	CDH26.7	18501	818	1368	2282	3013	2352	1668	1426	1277	1239	1280	1053	725
(14) Wind	WSAvg	0.4	0.3	0.4	0.4	0.5	0.5	0.4	0.3	0.3	0.3	0.3	0.3	0.3
(15) Precipitation	PrecAvg	1311	5	12	41	54	188	160	180	181	277	173	31	8
	PrecMax	1660	36	56	123	138	380	253	301	346	452	345	104	56
	PrecMin	829	0	0	0	0	51	51	65	86	144	16	1	0
	PrecStd	226	9	15	34	38	95	53	65	58	80	82	31	15
	PrecMax	1660	36	56	123	138	380	253	301	346	452	345	104	56
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	36.7	38.2	40.4	39.8	36.7	35.9	34.8	34.8	34.7	34.8	34.2
		MCWB	23.4	24.2	24.2	25.1	26.0	26.5	26.3	26.5	26.7	26.6	25.6	24.3
	2%	DB	33.4	35.5	37.1	38.7	37.8	35.0	34.3	33.5	33.7	33.6	33.8	33.0
		MCWB	23.2	24.0	24.6	25.7	26.3	26.7	26.4	26.6	26.7	26.4	25.5	24.2
	5%	DB	32.4	34.5	36.1	37.5	36.3	33.8	33.1	32.6	32.8	32.9	32.7	31.9
		MCWB	23.0	24.0	24.5	26.0	26.5	26.7	26.4	26.5	26.6	26.2	25.1	23.7
	10%	DB	31.2	33.4	35.0	36.4	34.8	32.8	32.1	31.7	31.9	32.0	31.7	30.6
		MCWB	22.8	23.9	24.6	26.1	26.5	26.6	26.3	26.4	26.6	26.1	24.8	23.2
	0.4%	WB	25.7	27.6	27.8	28.7	28.6	28.4	27.7	27.7	28.0	27.7	27.2	26.6
		MCDB	31.4	34.2	34.3	35.7	34.8	33.3	32.2	32.0	32.1	32.0	31.9	31.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	25.0	26.3	26.7	27.7	27.8	27.8	27.2	27.2	27.5	27.2	26.5	25.5
		MCDB	30.3	32.6	33.6	35.0	33.5	32.5	31.5	31.4	31.6	31.4	31.3	30.3
	5%	WB	24.4	25.6	26.2	27.2	27.3	27.4	27.0	26.9	27.1	26.9	26.0	24.7
		MCDB	29.5	31.7	33.0	34.4	32.9	31.8	31.2	31.0	31.2	31.0	30.7	29.6
	10%	WB	23.7	25.0	25.7	26.7	27.0	27.0	26.7	26.6	26.7	26.6	25.5	24.1
		MCDB	28.8	30.8	32.3	33.6	32.4	31.2	30.5	30.4	30.6	30.4	29.9	28.8
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	12.4	11.9	10.9	8.7	7.3	6.9	6.8	7.2	7.6	9.2	10.8
		MCDBR	13.3	13.8	13.5	12.5	10.7	8.6	8.4	8.0	8.4	8.9	10.5	11.6
		MCWBR	4.7	4.6	3.9	3.1	2.6	2.5	2.6	2.5	2.7	2.9	3.4	3.9
	5% WB	MCDWR	10.7	11.6	11.4	10.8	9.0	7.6	7.2	7.2	7.6	8.0	9.2	10.2
		MCWBR	4.3	4.6	3.9	3.5	2.9	2.7	2.6	2.5	2.8	2.9	3.3	3.8
(40) Clear-Sky Solar Irradiance	taub		0.481	0.573	0.679	0.616	0.493	0.436	0.428	0.432	0.468	0.550	0.479	0.479
	taud		2.085	1.868	1.651	1.838	2.221	2.399	2.421	2.394	2.281	1.999	2.178	2.109
	Ebn at Noon		803	747	683	730	814	854	862	866	839	755	801	791
	Edh at Noon		156	203	259	215	145	120	117	122	136	175	141	149
	RadAvg		4.63	5.18	5.42	5.93	5.52	4.99	4.69	4.61	4.76	4.59	4.67	4.46
(45) All-Sky Solar Radiation	RadStd		0.23	0.21	0.36	0.32	0.40	0.39	0.32	0.26	0.27	0.29	0.28	0.18

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days		
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	+0.13	N/A	N/A	+107

Nomenclature: See separate page